

that can reach a length of 3 m. It hunts deer and other large prey, delivering venom with its bite.

Snakes descended from lizards with legs (see Figure 26.1); thus they are classified as legless lizards. Today, some species of snakes retain vestigial pelvic and limb bones, providing evidence of their ancestry. Despite their lack of legs, snakes are quite proficient at moving on land, most often by producing waves of lateral bending that pass from head to tail. Force exerted by the bends against solid objects pushes the snake forward. Snakes can also move by gripping the ground with their belly scales at several points along the body while the scales at intervening points are lifted slightly off the ground and pulled forward.

Snakes are carnivorous, and a number of adaptations aid them in hunting and eating prey. They have acute chemical sensors, and though they lack eardrums, they are sensitive to ground vibrations, which helps them detect the movements of prey. Heat-detecting organs between the eyes and nostrils of pit vipers, including rattlesnakes, are sensitive to minute temperature changes, enabling these night hunters to locate warm animals. Venomous snakes inject their toxin through a pair of sharp teeth that may be hollow or grooved. The flicking tongue is not venomous but helps fan odors toward olfactory (smell) organs on the roof of the

mouth. Loosely articulated jawbones and elastic skin enable most snakes to swallow prey larger than the diameter of the snake's head (see Figure 23.14).

Next, we'll turn to the third (and final) clade of reptiles with living members, the archosaurs. There are two extant groups of archosaurs: the crocodilians and the birds.

Crocodilians

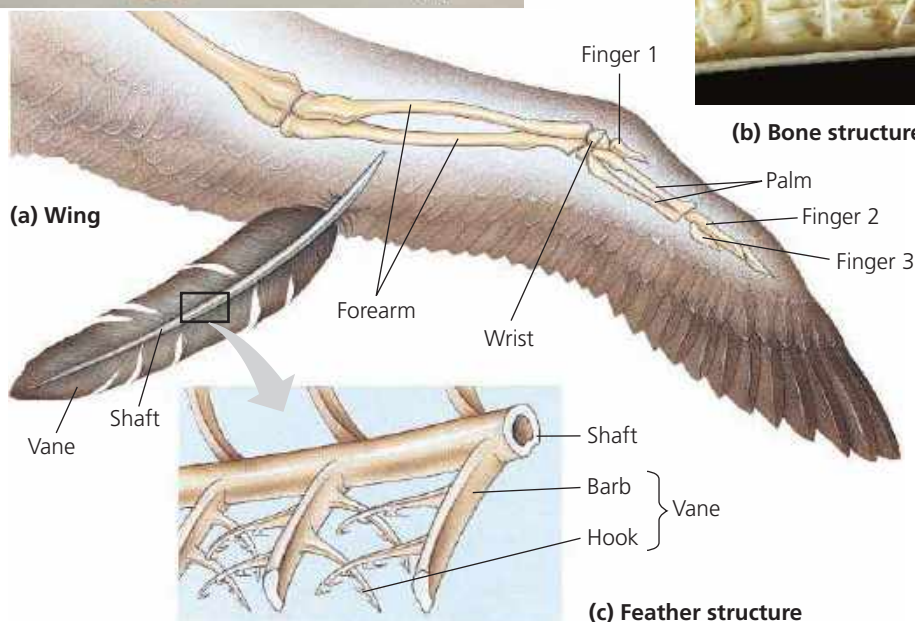
Alligators and crocodiles (collectively called crocodilians) belong to a lineage that reaches back to the late Triassic. The earliest members of this lineage were small terrestrial quadrupeds with long, slender legs. Later species became larger and adapted to aquatic habitats, breathing air through their upturned nostrils. Some Mesozoic crocodilians grew as long as 12 m and may have attacked dinosaurs and other prey at the water's edge.

The 24 known species of living crocodilians are confined to warm regions of the globe. In the southeastern United States, the American alligator (**Figure 34.29e**) has made a comeback after spending years on the endangered species list.

Birds

There are about 10,000 species of birds in the world. Like crocodilians, birds are archosaurs, but almost every feature of their anatomy has been modified in their adaptation to flight (**Figure 34.30**).

Derived Characters of Birds Many of the characters of birds are adaptations that facilitate flight, including



(b) Bone structure

(c) Feather structure

◀ Figure 34.30 Form fits function: the avian wing and feather. (a) A wing is a remodeled version of the tetrapod forelimb. (b) The bones of many birds have a honeycombed internal structure and are filled with air. (c) A feather consists of a central air-filled shaft, from which radiate the vanes. The vanes are made up of barbs, which bear small branches called barbules. Birds have contour feathers and downy feathers. Contour feathers are stiff and contribute to the aerodynamic shapes of the wings and body. Their barbules have hooks that cling to barbules on neighboring barbs. When a bird preens, it runs the length of each contour feather through its beak, engaging the hooks and uniting the barbules into a precise shape. Downy feathers lack hooks, and the free-form arrangement of their barbules produces a fluffiness that provides insulation by trapping air.

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